

Multi-physics analysis of a supercritical water reactor with improved MCNP modeling



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HIGHLIGHTS

- An example SCWR assembly was analyzed with MCNP and a custom-developed single-pin-cell code.
- Thermal scattering effects and MAKXSf-derived cross-sections were included in the reactor physics analysis.
- A significant reactivity effect ($\$2$) was noted between this detailed treatment and conventional (Doppler broadening-only) analysis.
- Neutron flux profiles are also changed significantly based on a more detailed reactor physics treatment.

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ABSTRACT

Supercritical water reactors (SCWR) are proposed Generation IV light water reactors where supercritical water acts as the working fluid. Due to the large density gradients in the coolant, evaluation of reactor designs requires a detailed multi-physics analysis. In this work, MCNP is coupled to a single channel analysis tool to analyze a fuel assembly design whose neutronics performance has been experimentally validated. Accurate modeling of the reactor physics in a SCWR requires a detailed temperature treatment, including the effects of Doppler broadening (TMP cards in MCNP), thermal scattering (MT cards), and fully temperature-dependent cross-section sets for fuel and coolant (using the MAKXSf code). There is a $\$2$ difference in k_{eff} between this highest-fidelity approach and a conventional treatment that addresses only Doppler broadening. An analysis is performed to determine the size of the temperature intervals for custom-developed cross-section sets, which indicated the necessity of 50 K bins in the fuel and 10 K bins in the coolant.

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1. Introduction

Supercritical water reactors (SCWR) are being examined as one of the Generation IV designs due to the simplification of the balance of plant and the increased thermal efficiency of the supercritical Rankine cycle (Fig. 1). Operating experience for the supercritical Rankine cycle has been obtained in coal-fired power plants. The conditions present in supercritical water systems can be seen in Fig. 1.

One design is the High Performance Light Water Reactor (HPLWR), a thermal SCWR to output 1000 MWe. The fuel assemblies consist of 40 fuel pins in a square lattice surrounding a central moderator box (Fig. 2a). Research and development for this design

has included reactor safety assessments, exploration of core geometry parameters, preliminary materials selection, and economic assessments (Squarer et al., 2003).

The Super Light Water Reactor (SLWR) is a thermal SCWR designed to output 1200 MWe. The fuel assemblies each include 300 fuel pins, a central flux monitor guide tube, and 36 moderator boxes. Fig. 2(b) shows the structure of this design (Kamei et al., 2002). Fig. 3 shows a sample lattice.

In all LWRs, there is a strong coupling of the reactor physics and thermal hydraulics due to the use of water as both coolant and moderator. The reactor physics calculations determine the heat source that the thermal hydraulic calculations receive as input. The resulting profiles of coolant density and temperature provide feedback to the reactor physics calculations. While the feedback for operating subcritical LWRs is adequately understood, additional study is required to extend these analyses to the SCWR.

Density variations in pressurized water reactors are continuous but are limited to a small range, allowing for linear approximations.

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Nomenclature

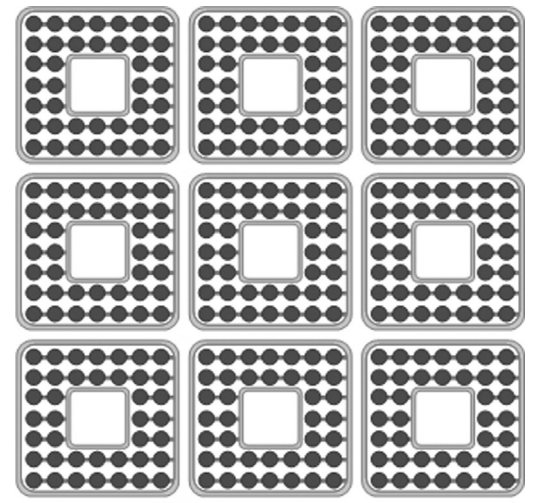
C_p	constant pressure specific heat [$\text{J kg}^{-1} \text{K}^{-1}$]
D	diameter [m]
f	friction factor
h	specific enthalpy [J kg^{-1}]
htc	heat transfer coefficient [$\text{W m}^{-2} \text{K}^{-1}$]
k	thermal conductivity [$\text{W m}^{-1} \text{K}^{-1}$]
k_{eff}	neutron production to loss ratio
Nu	Nusselt number
P	pitch [m]
Pr	Prandtl number
q'	linear heat rate [W m^{-1}]
Re	Reynolds number
T	temperature [K]

Greek symbols

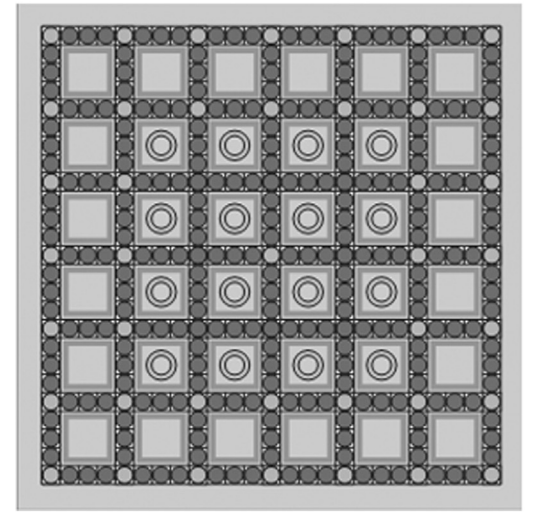
β_{eff}	delayed neutron fraction
Δp	pressure loss [Pa]
Δz	fuel element height [m]
ρ	density [kg m^{-3}]
σ	Stefan–Boltzmann constant [$\text{W m}^{-2} \text{K}^{-4}$]
ψ	geometric correction factor

Subscripts

b	bulk
ci	inner clad
f	fuel
fo	outer fuel
gap	gas gap
i	iteration
p	pin number
pin	fuel pin
$r.t.$	round tube
w	wall
z	axial location



(a) HPLWR



(b) SLWR

Fig. 2. SCWR designs.

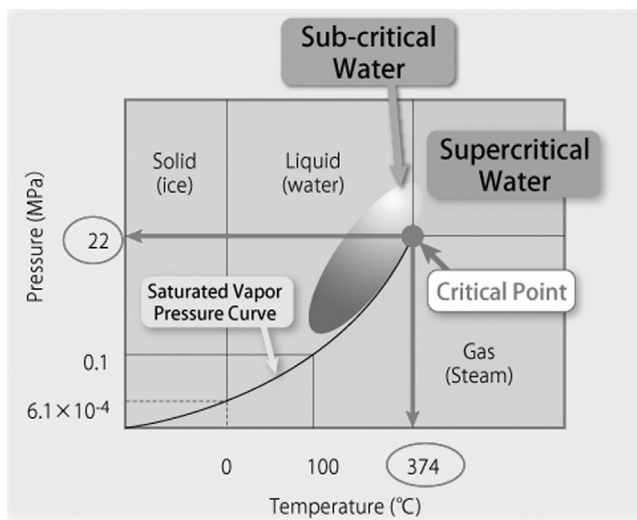


Fig. 1. Phase diagram of water (AsiaBiomassOffice, 2013).

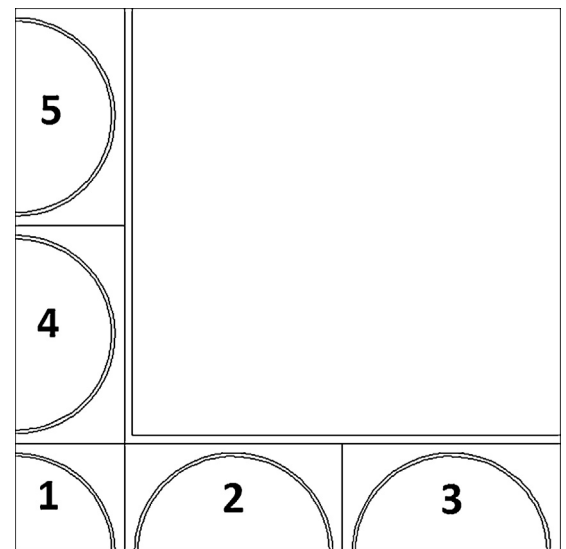


Fig. 3. Pin numbering scheme.

Density variations in boiling water reactors are larger but discontinuous, allowing the use of void fraction models. In SCWRs, density variations are both large and continuous, requiring the use of density distribution in reactor physics calculations.

Yamaji performed a 3D multi-physics coupling for a SCWR fuel assembly (Yamaji et al., 2005). Yamaji coupled SRAC, a reactor physics analysis code developed at JAERI, with SPROD, a single channel analysis tool developed at the University of Tokyo. A major new result was a calculated, average coolant core outlet temperature. SPROD passed local water density to SRAC in 40 axial cells, but did not address temperature variation in coolant, cladding, and fuel, or the resulting effects on cross-sections.

Wataa coupled MCNP with STAFAS, a single channel analysis tool designed for use in SCWR fuel designs (Wataa, 2005). Temperature effects in the reactor physics are considered through interpolation of different cross-section libraries to mimic unique temperature cross-sections for each volume element. Wataa used TMP cards in MCNP to specify the temperature of each volume element which adjusted the elastic scattering cross-section of the material.

Wataa did not include a detailed thermal scattering treatment (through MCNP's MT cards). This detailed treatment is often necessary, particularly for water, due to $S(\alpha, \beta)$ scattering interactions. Neglecting thermal scattering effects will underestimate the neutron scattering in a material.

Wataa considered the fuel conductivity to be constant and used the Bishop correlation (Bishop et al., 1964) to determine the heat transfer at the wall.

$$Nu = 0.00459 Re^{0.923} Pr^{0.613} \left(\frac{\rho_w}{\rho_b} \right)^{0.231} \quad (1)$$

An axial resolution of 21 cells was used to model the system.

Chaudri et al. coupled MCNP with SACoS, another SCWR sub-channel analysis tool, to model the reference HPLWR fuel assembly (Chaudri et al., 2012). Similar to Wataa, Chaudri accounts for temperature effects in reactor physics calculations through interpolation of the cross-section libraries. The analysis of conduction heat transfer in the pin and of flow boundary conditions was not discussed, nor were experimental data considered.

Xi et al. (2013) employed a detailed CFX model coupled to MCNP. However, their paper does not discuss the reactor physics calculation methods in depth, with no information on how temperature-dependence was addressed.

An MCNP model, validated by integral experiments, is used in this work to confirm the reliability of the neutronics models. A prototype SCWR fuel assembly was loaded into the central experimental region of the PROTEUS Research Reactor (Rätz, 2012). The fuel assembly inserted into the PROTEUS reactor was based on a design that is similar to the SLWR design but with a larger fuel pin diameter (0.0122 m) and pitch (0.0134 m). Fig. 4 is the full core and Fig. 5 is a reduced lattice for the PROTEUS Research Reactor. Since the integral experiments demonstrate the validity of MCNP results with this configuration, the model was based around this design.

The total fission rate and the ^{238}U capture rate were measured to determine the spatial neutron flux distribution within the fuel region through a gamma scanning technique. MCNPX simulations agreed with the total fission rates measured to within 0.4% (Rätz, 2012), similar to a 1σ uncertainty. Fig. 6 shows one of the validation results obtained at PROTEUS, which indicates good agreement for all pins (within 1σ for 59/61). This demonstrates that MCNP can handle the density variations that are expected to occur within an SCWR (Rätz et al., 2012).

This analysis includes the following improvements to the above-cited works:

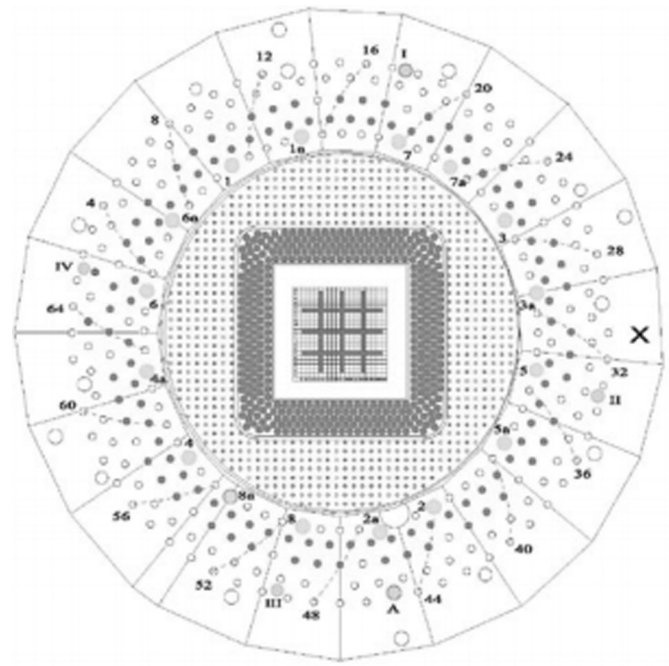


Fig. 4. PROTEUS diagram with outside graphite driver rings, D₂O driver region, the square natural uranium buffer zone, and the test tank (Chawla et al., 2013).

1. TMP cards are used to adjust the elastic scattering cross-sections in the fuel, cladding, and coolant.
2. Axial resolution is increased (80 axial zones for 4.2 m).
3. A temperature dependent fuel conductivity correlation is implemented.
4. A heat transfer correlation that considers heat transfer deterioration is used.
5. MT cards are employed to address $S(\alpha, \beta)$ interactions (thermal scattering treatment).
6. The temperature bins for cross-section evaluation are varied to ensure an accurate, efficient solution within memory limitations.

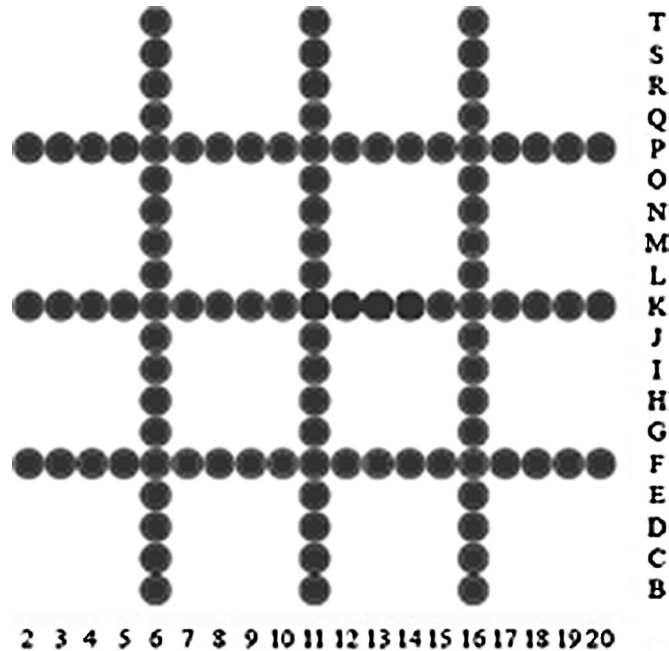


Fig. 5. Reduced SCWR lattice (Chawla et al., 2013).

Q						1.008 0.005							
P		0.995 0.005	0.997 0.004	1.010 0.004	1.001 0.004	0.997 0.004	0.998 0.005	0.999 0.004	0.994 0.004	1.003 0.004	1.008 0.005	0.997 0.005	
O		1.003 0.004					1.003 0.005					0.995 0.004	
N		0.997 0.004					1.004 0.004					1.005 0.004	
M		1.002 0.004					0.997 0.005					0.999 0.004	
L		0.999 0.004					1.005 0.004					1.002 0.005	
K	0.994 0.004	0.998 0.005	1.000 0.004	1.003 0.004	1.002 0.004	1.001 0.004	0.996 0.004	1.000 0.004	0.991 0.004	1.004 0.004	1.000 0.004	0.995 0.005	0.997 0.004
J		1.000 0.004					1.006 0.005					0.999 0.004	
I		0.996 0.004					0.997 0.005					1.003 0.005	
H		1.001 0.004					1.002 0.004					0.999 0.004	
G		0.997 0.004					0.996 0.005					1.003 0.004	
F		1.003 0.004	0.998 0.004	0.998 0.004	1.002 0.004	0.999 0.004	1.000 0.005	1.000 0.004	1.001 0.004	1.002 0.004	0.996 0.004	0.995 0.005	
E							1.000 0.005						
	5	6	7	8	9	10	11	12	13	14	15	16	17

Fig. 6. Calculation/experiment (C/E) values for the normalized distribution of the total fission rate. The lower value is the 1σ uncertainty. Highlighted areas have greater than 1σ deviation between experiment and calculations (Rätz et al., 2011).

2. Coupling

2.1. Initialization

A constant axial temperature gradient for all pins is used in iteration 0 of the single channel analysis (SCA) tool. The results are then used for the first MCNP run.

2.2. MCNP

The temperature of each cell (coolant, cladding, fuel) and the density of each coolant cell are provided to the MCNP model used in the coupling. Temperatures are used to adjust cross-sections, as detailed in Section 3. After the simulation, the fission energy deposition (F7 tally) for each fuel cell is recorded.

2.3. Post-processing

The fission energy deposition tallies from MCNP are normalized to maintain total power between iterations. This maintains constant inlet and outlet coolant temperatures and total mass flow rate at the values expected in the proposed SCWR assembly.

2.4. Single channel analysis tool

The SCA receives the linear heat profile and calculates the temperature and density in each coolant cell by adding the heat

generated in the current cell to the enthalpy of the previous cell.

$$h_{p,z} = h_{p,z-1} + q'_{p,z} \Delta z \quad (2)$$

$$T_{p,z} = T(h_{p,z}, p) \quad (3)$$

$$\rho_{p,z} = \rho(T_{p,z}, p) \quad (4)$$

2.5. Convergence

For convergence to be confirmed, two criteria must be met in each cell.

1. Coolant density change must be below 5% between iterations.
2. Linear heat profile change must be below 5% between iterations

Density is normalized based upon the current cell's value. Linear heat is normalized by the maximum linear heat rate.

$$0.05 > \frac{|\rho_i - \rho_{i-1}|}{\rho_i} \quad (5)$$

$$0.05 > \frac{|q'_i - q'_{i-1}|}{q'_{\max,i}} \quad (6)$$

If both criteria are not met, the SCA provides the MCNP component with the most recent temperatures and densities and the process repeats.

Table 1

Dimensions and materials of SCWR model lattice.

Fuel pellet diameter	0.00878 m
Clad thickness	0.00071 m
Pitch	0.0112 m
Active fuel height	4.27 m
Moderator box wall thickness	0.0004 m
Coolant flow rate	1843 kg s ⁻¹
Fuel type	UO ₂ fuel pellet
Fuel density	10412.5 kg m ⁻³
Clad/water box material	D9 alloy
Clad/water box density	8100 kg m ⁻³
Coolant/moderator	H ₂ O

3. Reactor physics modeling

MCNP5 (X-5 Monte Carlo Team, 2003) was used to simulate the SCWR test lattice seen in Fig. 3. A quarter assembly was modeled due to symmetry. The fuel pins were modeled as fuel and clad only. The gas gap is neglected in order to optimize the MCNP calculation. The additional fuel in the gas gap is less than 0.04% of the fuel pellet. Table 1 provides the dimensions for the lattice. The model was divided into 80 axial levels of equal height. Reflective boundaries were used in the x and y directions with vacuum boundaries in the z direction.

The lattice model included 3 materials: UO₂ fuel, D9 alloy (for cladding and water box), and water. Table 2 shows the non-iron composition of D9 used in MCNP. The fuel enrichment for the model (2.38 at%) was chosen so that the lattice was near critical. This selected enrichment simulated operating conditions and minimized errors associated with modeling a reactor with reactivity in excess of \$1.

To determine what simplifications were acceptable in the reactor physics calculation for an SCWR, the first set of iterations were based on the simplest neutronics model. With each subsequent group, a level of detail was added to the reactor physics by modifying the available cross-sections. All calculations were performed using 50,000 source histories per cycle, 1000 cycles, skipping the first 500 cycles, and a guess k_{eff} value of 1.0. The first group, “Room Temp XS”, used cross-sections at room temperature for all materials. Only TMP cards were employed, with no thermal scattering treatment (MT cards). TMP cards only modify the elastic scattering cross-sections. Total Doppler broadening is only possible through the use of temperature dependent cross-sections.

In the next group, “Global Temp XS”, cross-sections were chosen from the default MCNP libraries that were most consistent with thermal hydraulic results from “Room Temp XS”: 600 K for the cladding, 600 K for water, and 900 K for the fuel. Again, Doppler broadening was addressed, but not thermal scattering.

In the subsequent group, “Global Temp XS/MT”, thermal scattering was address through MT cards in the MCNP input. There were three sub groups within “Global Temp XS/MT”: “Global Temp XS/MT(Fuel)”, with MT cards corresponding to the fuel (800K O,

Table 2

D9 alloy composition (in wt.% – Fe balance).

Ni	15.5%
Cr	14.04%
Mo	2.25%
Mn	1.78%
Si	0.61%
Ti	0.26%
C	0.04%
P	0.007%
Al	0.006%
N	0.0037%
S	0.002%
B	0.0015%

800K U²³⁸), “Global Temp XS/MT(Water)”, corresponding to the coolant and water in the moderator box (650 K and 550 K light water, respectively), and “Global Temp XS/MT(Both)” which combined both effects.

The final group, “Binned Temp XS/MT”, used cross-sections developed through the MAKXSF code packaged with MCNP. MAKXSF calculates cross-sections at any temperature. Cells with a temperature close to the MAKXSF evaluated temperature were assigned the corresponding cross-section. ENDF/B-VII cross-sections have a temperature interval of 300 K. Smaller temperature intervals allow for a more accurate representation of the cross-sections at the material’s temperature. However, finer discretization increases memory usage, requiring an optimization between accuracy and computational resources. The temperature bins tested were 50, 100, and 150 K for the fuel; 25, 50, and 100 K for the cladding; and 5, 10, 25, 50, and 100 K for the coolant.

The coolant densities varied axially based on feedback from the thermal hydraulics. The density of the water in the moderator box was constant and evaluated at 563 K and 25 MPa. This is the average temperature of the moderator (inlet of 553 K, outlet of 573 K) (Buongiorno and MacDonald, 2003).

The fuel pins, seen in Fig. 3, were classified into 3 types: corner pin, the bottom left quarter pin (pin 1); middle pin, the two half pins next to the corner pin (pins 2 and 4); and end pin, the two half pins located at the bottom right and top left (pins 3 and 5). The MCNP results in Fig. 7(a) and (b) verify the symmetry in this lattice.

The MCNP model received coolant densities and temperature distributions of the fuel, cladding, and coolant from the thermal model. The MCNP model reported the power production in each cell.

4. Thermal hydraulic modeling

Thermo-physical property data provided by NIST (2011) were used to create a properties table. A pressure of 25 MPa was used with 0.01 K increments from 553 K (inlet) to 830 K. The upper limit was set to allow use of the Swenson heat transfer correlation (Swenson et al., 1965), which includes wall temperature effects.

$$Cp_0 = \frac{h_w - h_b}{T_w - T_b} \quad (7)$$

$$Nu = 0.00459 Re_w^{0.92} Pr_w^{0.61} \left(\frac{\rho_w}{\rho_b} \right)^{0.23} \left(\frac{Cp_0}{Cp_w} \right)^{0.61} \quad (8)$$

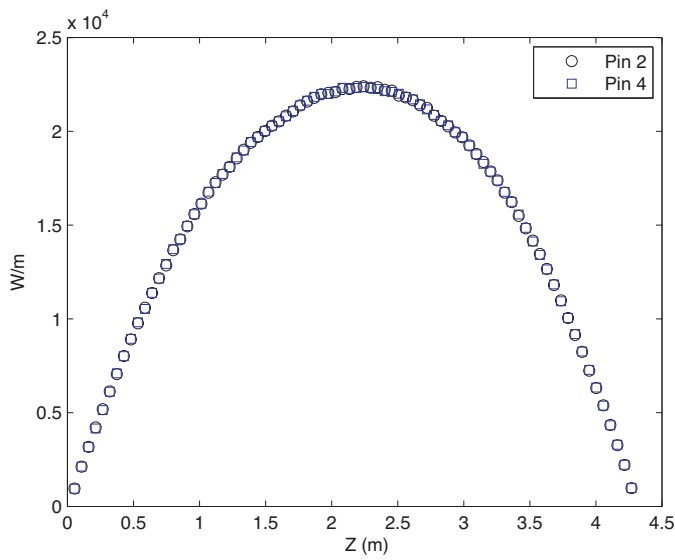
The solver divided the SCWR geometry into 80 axial segments to match the MCNP model. Each segment had a corresponding linear heat rate from MCNP. The solver determined the temperatures of the coolant, the outer and inner walls of the clad, and the surface and center of the fuel element as follows.

The inlet temperature, inlet pressure, and outlet pressure were kept equal for all five channels. To ensure outlet pressure is equal, pressure loss was calculated. The friction factor was calculated using a suggested equation for supercritical water (Petrov and Popov, 1988).

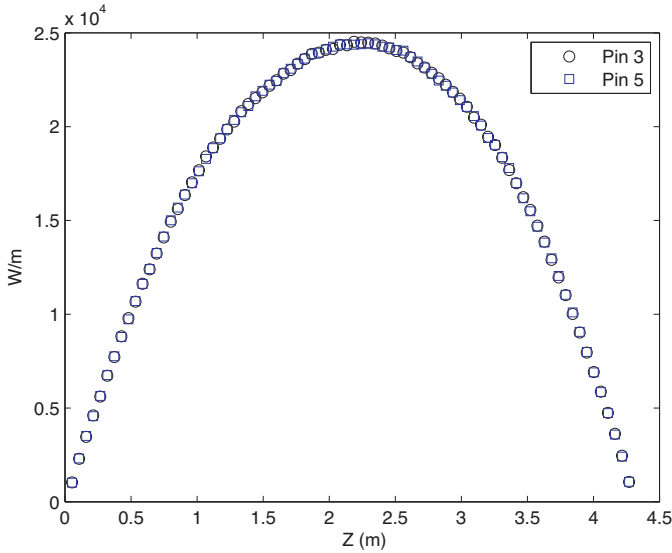
$$f = \left(1.82 \log \left(\frac{Re}{8} \right) \right)^{-2.0} \left(\frac{\rho_w}{\rho_b} \right)^{0.4} \quad (9)$$

Gravity and acceleration terms are also employed. The average mass flow rate per channel was assumed equal for all channels as an initial guess. The mass flow rate was adjusted for each channel until the pressure difference among all channels was less than 100 Pa (0.5% of the total pressure loss).

The Swenson heat transfer coefficient was solved iteratively, with Dittus–Boelter as an initial guess. The correlations were



(a) Middle pins



(b) End pins

Fig. 7. Linear heat profile comparison.

modified to reflect the fuel assembly geometry using a correction factor (Presser, 1967).

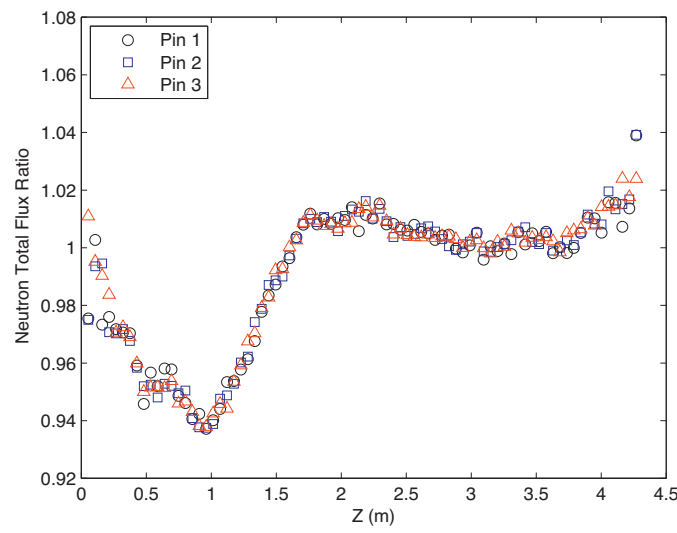
$$\psi = 0.9217 + 0.1478 \frac{P}{D_{pin}} - 0.1130e^{-7(P/D_{pin}-1)} \quad (10)$$

$$htc_{pin} = \psi htc_{r.t.} \quad (11)$$

The Swenson correlation determined the outer clad temperature. Conduction through D9 provided the inner clad temperature. The thermal conductivity of D9 at 650 K, the average clad temperature, is $18.9 \text{ W m}^{-1} \text{ K}^{-1}$ (Leibowitz and Blomquist, 1988). An open helium gap determined the temperature difference between the inside of the cladding and outside of the fuel. An empirical correlation estimated the gas thermal conductivity (in $\text{W m}^{-1} \text{ K}^{-1}$) (Von Ubusch et al., 1958).

$$k_{gap} = 15.8 \times 10^{-4} \left(\frac{T_{ci} + T_{fo}}{2} \right)^{0.79} \quad (12)$$

$$htc_{gap} = \frac{k_{gap}}{(D_{ci} - D_{fo})/2} + \sigma \frac{T_{fo}^4 - T_{ci}^4}{T_{fo} - T_{ci}} \quad (13)$$

**Fig. 8.** Room Temp XS over Global Temp XS.

Todreas and Kazimi (1990) provided the method to compute heat transfer in the uranium oxide fuel. An iterative method was employed based on a correlation (Hann et al., 1973) for the conductivity integral:

$$\int k_f dT = 3824 \log(402.4 + T) + 1256 \times 10^{-11} \frac{(T)^4}{4} \quad (14)$$

The radial temperature distributions were averaged within each material to provide temperatures for the MCNP model.

5. Results

5.1. Room temp and global temp

MCNP calculated k_{eff} to be 1.01871 ± 0.00009 and 1.01011 ± 0.00009 for “Room Temp XS” and “Global Temp XS” respectively. There is a difference of 1.3\$ of reactivity, assuming β_{eff} of 650 pcm. Fig. 8 shows the ratio of the axial neutron flux from “Global Temp XS” to “Room Temp XS”. The difference in neutron flux, and thus q' , is as high as 6%. All three fuel pins exhibit similar behavior within a given group. As a result, the remaining analysis will focus on pin 2.

5.2. Thermal scattering (MT cards)

MCNP calculated k_{eff} to be 1.01099 ± 0.00009 , 0.99892 ± 0.00009 , and 1.00002 ± 0.00009 for “Global Temp XS/MT(Fuel)”, “Global Temp XS/MT(Water)”, and “Global Temp XS/MT(Both)”, respectively.

A detailed thermal scattering treatment in fuel shows a smaller effect on k_{eff} (0.14\$) than in water (1.72\$), since thermal scattering is a more significant effect in water. Fig. 9 shows the ratio of the axial neutron fluxes of the thermal scattering models with respect to “Global Temp XS”. The figure shows that without thermal scattering, the model overestimates the power in the bottom half of the reactor and underestimates the peak power.

Thermal scattering in water accounts for about 90% of the total thermal scattering effect in power profile and k_{eff} . Since inclusion of thermal scattering effects comes at appreciable computational expense, particularly for system memory, it was determined unnecessary for the fuel. Fig. 10 shows the ratio of the thermal epithermal, and fast fluxes in each cell between “Global Temp XS/MT(Water)” and “Global Temp XS/MT(Both)”. This shows that

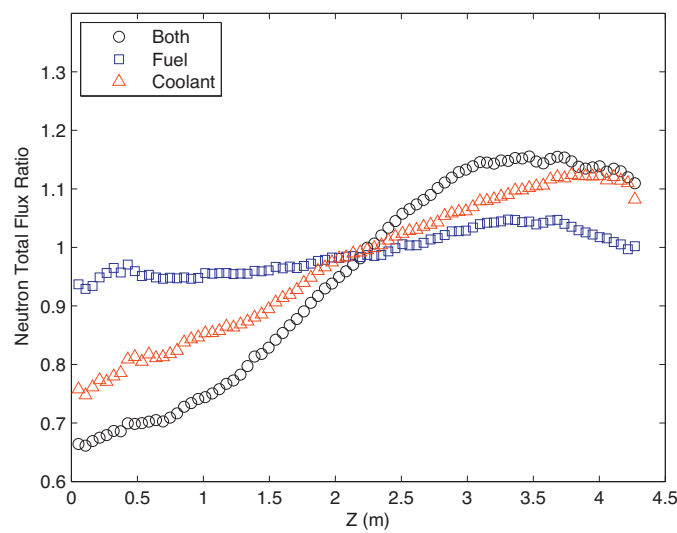


Fig. 9. Pin 2 Global Temp XS/MT over Global Temp XS.

neglecting thermal scattering in the fuel does not cause a significant shift in the predicted neutron spectrum.

5.3. Temperature bins

New cross-sections are created by interpolating between old cross-sections through MAKXSf. The new cross-sections represent the material at a set temperature. All materials with a temperature similar are grouped into a temperature bin and are modeled with the same temperature cross-section. Smaller bins require more cross-sections to be created.

MCNP calculated k_{eff} to be 0.99623, 0.99627, and 0.99621 for the 50, 100, and 150 K bins in fuel. k_{eff} was 0.99935, 0.99920, and 0.99905 for the 25, 50, and 100 K bins in cladding. k_{eff} was 1.00025, 1.00006, 1.00031, 1.00026, and 1.00045 in the 5, 10, 25, 50, and 100 K bins in coolant. All k_{eff} values have a standard deviation of 0.00009.

There is little reactivity difference between temperature bins within each material. Fig. 11 shows the ratio of the flux profiles for each temperature bin in the fuel sub group with respect to “Global Temp XS/MT(Water)”. This shows the need for temperature binning in the fuel as there can be a 60% difference in power production.

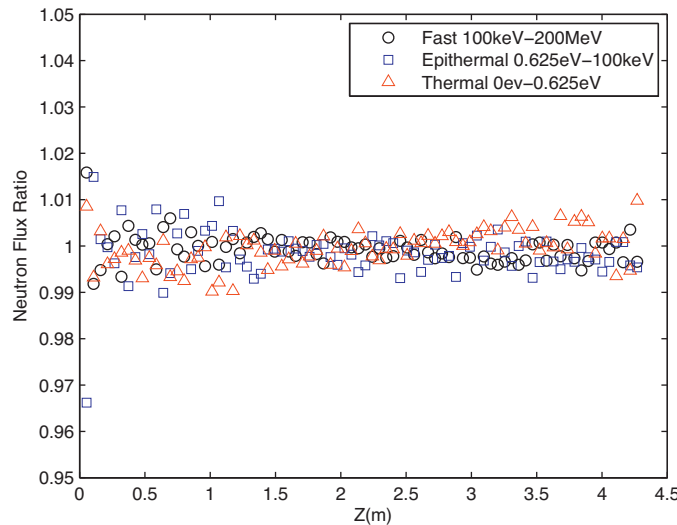


Fig. 10. Pin 2 Global Temp XS/MT(Water) over Global Temp XS/MT(Both).

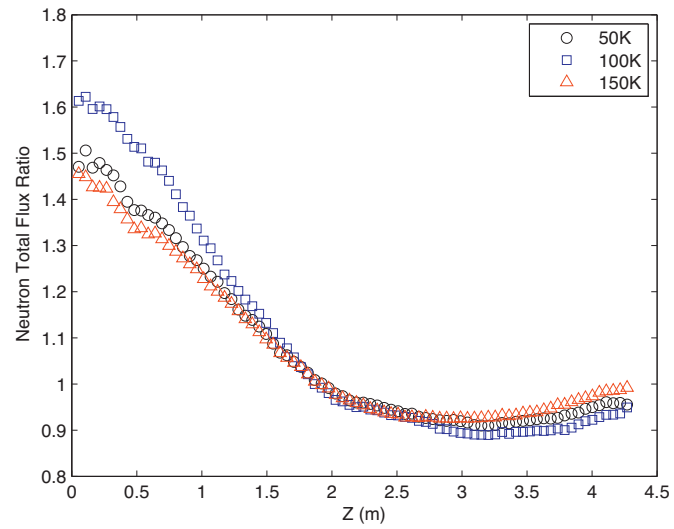


Fig. 11. Pin 2 fuel Binned Temp XS/MT variation.

Since the fuel determines power production, the highest accuracy is needed. Therefore, the 50 K fuel bins are necessary.

Fig. 12 shows the ratio of the flux profiles for each temperature bin in the clad sub group with respect to “Global Temp XS/MT(Water)”. There is a trend toward unity as the temperature bins get smaller. This is caused by a reduction in the magnitude of the jump discontinuities that exist when transitioning from one bin to another. Due to memory limits, the cladding is not binned. Fig. 13 shows the typical difference between 25 K cladding bins and “Global Temp XS/MT(Water)” is less than 2% with regards to the neutron flux. Because of the minimal differences in reactivity and linear heat, noting the clad thickness being less than 1% of the mean free path of a thermal neutron through D9 stainless steel, no temperature binning in the cladding was deemed an acceptable approximation.

Fig. 14 shows the ratio of the flux profiles for each coolant temperature bin width with respect to “Global Temp XS/MT(Water)”. Failure to bin coolant temperatures can underestimate the power production by 40% and overestimate it near the peak by 10%. Bins of 10 K were selected as a trade-off between accuracy and computational expense.

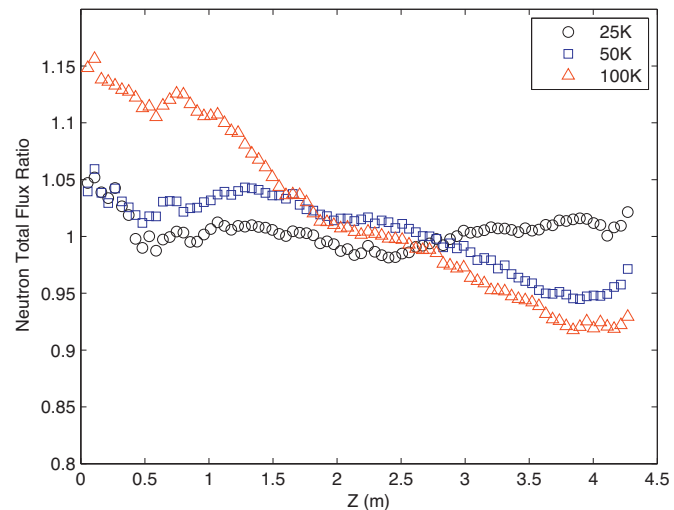


Fig. 12. Pin 2 clad Binned Temp XS/MT variation.

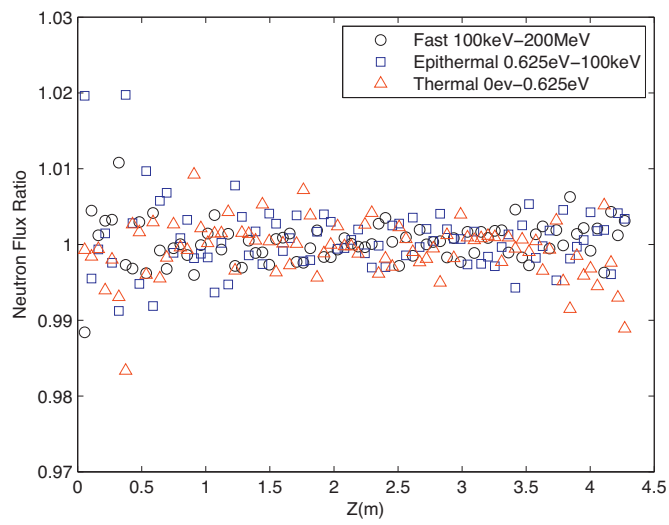


Fig. 13. Pin 2 clad Binned Temp XS/MT(25 K) over Global Temp XS/MT(Water).

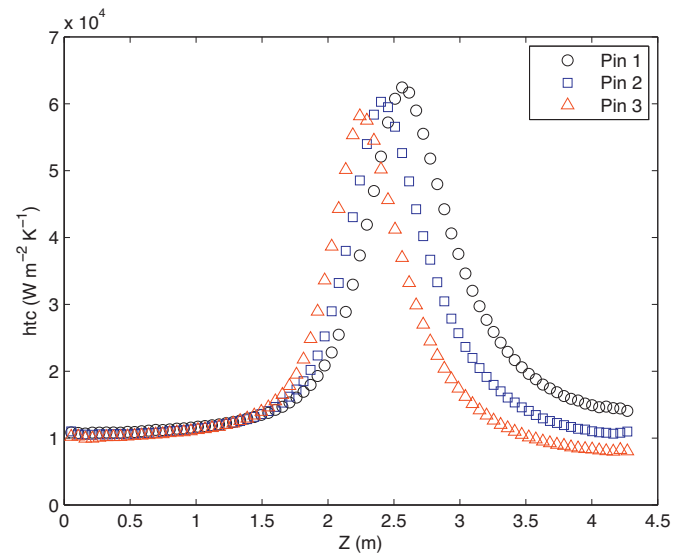


Fig. 16. Heat transfer coefficient.

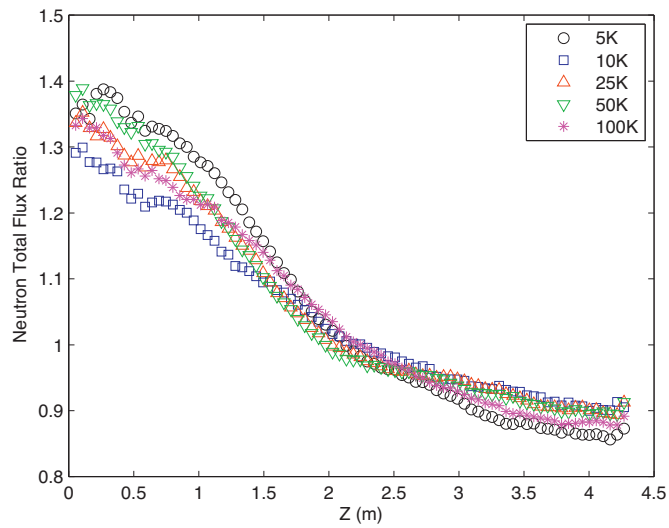
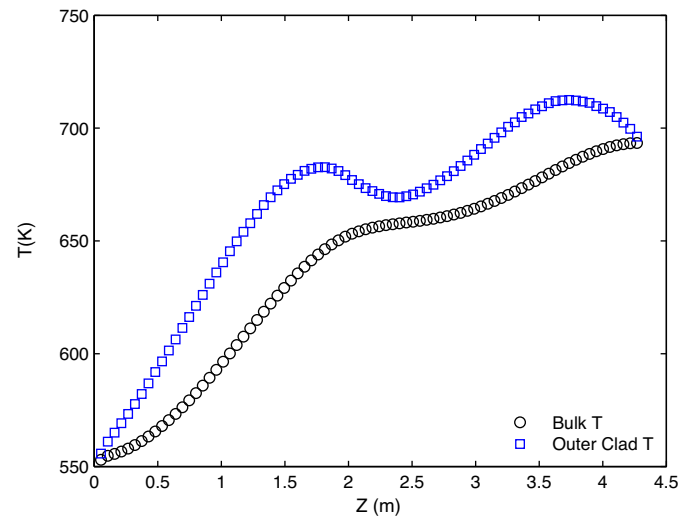


Fig. 14. Pin 2 coolant Binned Temp XS/MT variation.



(a) Coolant temperatures

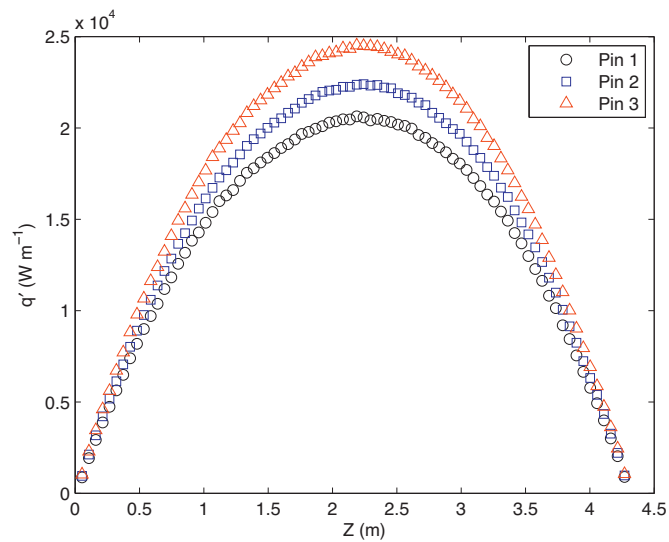
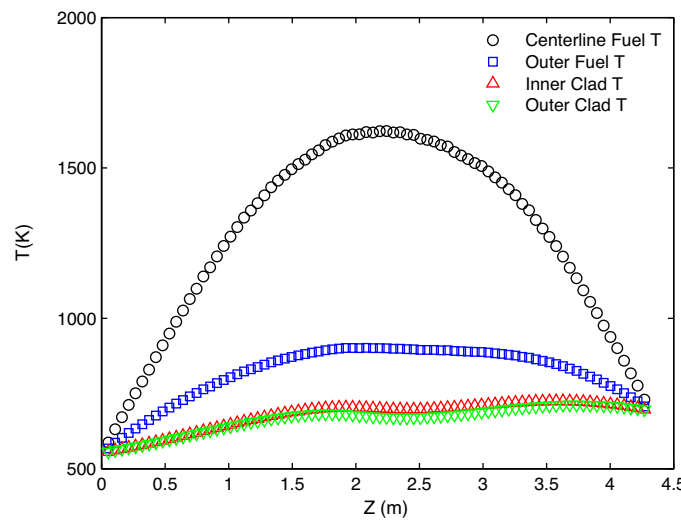


Fig. 15. Linear heat.



(b) Fuel rod temperatures

Fig. 17. Pin 2.

5.4. Final results

The final model has 50K binning in fuel and 10K binning in coolant, with TMP cards in each cell. This most-detailed model accounts for temperature effects and molecular resonance in the moderator. An under-relaxation factor of 0.5 was utilized to ensure convergence, using the same technique as (Wataa, 2005 (who used a value of 0.2)). k_{eff} is 1.00586 ± 0.00009 and β_{eff} is 0.00685 ± 0.00013 .

Fig. 15 shows that the linear heat rates peak above the midplane, located at 2.1 m in contrast to LWRs, which peak below. The supercritical fluid heat transfer coefficient is shown in Fig. 16. The hotter pins experience the pseudo-critical point at a lower height, causing the heat transfer coefficient peak to be closer to the bottom of the pin at a lower maximum value.

The heat transfer effects can be seen in the temperature profiles for the bulk fluid and outer clad (Fig. 17a). The outer clad temperature decreases near the peak linear heat, and then increases as the heat transfer coefficient decreases. This is another SCWR phenomena contrary to LWR behavior, where the peak heating is closely correlated to the clad temperature profile. Fig. 17(b) shows the fuel rod temperature profiles.

6. Conclusions

To preserve the validity provided by the experiments, a model based on the fuel assembly tested at PROTEUS was made in MCNP, coupled to a custom-built thermal analysis tool. This tool allowed estimates of temperatures for all pin cell materials as a function of axial height.

The need for detailed temperature-dependent cross-sections is demonstrated. Temperature bins are selected to be 50 K and 10 K for the fuel and coolant, respectively, with no binning in the cladding. This bin scheme was selected by considering higher accuracy in the materials important to the neutron physics while not exceeding computational limitations. When comparing the final temperature-binned model with the room temperature model, the power profile and k_{eff} differed by 20% and 2\$, respectively.

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